A QUADRATIC ELEMENT OF THE SQUARE OF AN IDEAL OF LARGE COLENGTH IS A SQUARE UP TO A FACTOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file resolve.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction in the first case through the partial derivatives and the comparison of degrees inside the square of the ideal, the characteristic two analysis using derivations of the field to produce polynomials of the shape constant plus square in the ideal, and the final exclusion through the conormal module and a dimension count with an explicit coordinate change.

Editorial note. The label lemma-issquare names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source resolve.tex –label lemma-issquare –start 3315 –end 3398 –local-id 28070 returned RESULT=OK (L28070 free, occupancy FREE, public interval clear). Pre-assembly tag self-check: canonical tag 0BGD, not present in reference/public-items.jsonl (neither by tag_url nor by claim_key suffix).

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Leibniz rule for the partial derivatives acting on the square of an ideal. Standard calculus of partial derivatives in a polynomial ring: the derivative of an element of the square of an ideal lies in the ideal. Supplies the fact that the two partial derivatives of the given quadratic element lie in the ideal, which drives the first case distinction.
- Extension of a derivation of the field with nontrivial value. Algebra, the lemma that for an element which is not a p-th power there is a derivation taking a nonzero value on it, extended to the polynomial ring by sending the variables to zero. Supplies the derivation used in characteristic two to produce elements of the ideal of a controlled shape.
- Freeness of the conormal module of the auxiliary ideal. Standard description of the conormal module of an ideal generated by a regular sequence of two elements in a two dimensional situation. Supplies the conclusion that the inclusion of the auxiliary ideal in the given one is strict, which produces an element with a nonzero mixed coefficient.

Source reference. Original lemma 031W: sources/stacks-chow/tags.txt, source lines 3999–3999. Prerequisite pointer invoked at proof line(s) 3363; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let κ be a field. Let $I \subset \kappa[x, y]$ be an ideal. Let*

$$a + bx + cy + dx^2 + exy + fy^2 \in I^2$$

for some $a, b, c, d, e, f \in k$ not all zero. If the colength of I in $\kappa[x, y]$ is > 1, then $a + bx + cy + dx^2 + exy + fy^2 = j(g + hx + iy)^2$ for some $j, g, h, i \in \kappa$.

Proof. Consider the partial derivatives $b + 2dx + ey$ and $c + ex + 2fy$. By the Leibniz rules these are contained in I. If one of these is nonzero, then after a linear change of coordinates, i.e., of the form $x \mapsto \alpha + \beta x + \gamma y$ and $y \mapsto \delta + \epsilon x + \zeta y$, we may assume that $x \in I$. Then we see that $I = (x)$ or $I = (x, F)$ with F a monic polynomial of degree ≥ 2 in y. In the first case the statement is clear. In the second case observe that we can write any element in I^2 in the form

$$A(x, y)x^2 + B(y)xF + C(y)F^2$$

for some $A(x, y) \in \kappa[x, y]$ and $B, C \in \kappa[y]$. Thus

$$a + bx + cy + dx^2 + exy + fy^2 = A(x, y)x^2 + B(y)xF + C(y)F^2$$

and by degree reasons we see that $B = C = 0$ and A is a constant.

To finish the proof we need to deal with the case that both partial derivatives are zero. This can only happen in characteristic 2 and then we get

$$a + dx^2 + fy^2 \in I^2$$

We may assume f is nonzero (if not, then switch the roles of x and y). After dividing by f we obtain the case where the characteristic of κ is 2 and

$$a + dx^2 + y^2 \in I^2$$

If a and d are squares in κ, then we are done. If not, then there exists a derivation $\theta : \kappa \to \kappa$ with $\theta(a) \neq 0$ or $\theta(d) \neq 0$, see Algebra, Lemma 031W. We can extend this to a derivation of $\kappa[x, y]$ by setting $\theta(x) = \theta(y) = 0$. Then we find that

$$\theta(a) + \theta(d)x^2 \in I$$

The case $\theta(d) = 0$ is absurd. Thus we may assume that $\alpha + x^2 \in I$ for some $\alpha \in \kappa$. Combining with the above we find that $a + \alpha d + y^2 \in I$. Hence

$$J = (\alpha + x^2, a + \alpha d + y^2) \subset I$$

with codimension at most 2. Observe that J/J^2 is free over $\kappa[x, y]/J$ with basis $\alpha + x^2$ and $a + \alpha d + y^2$. Thus $a + dx^2 + y^2 = 1 \cdot (a + \alpha d + y^2) + d \cdot (\alpha + x^2) \in I^2$ implies that the inclusion $J \subset I$ is strict. Thus we find a nonzero element of the form $g + hx + iy + jxy$ in I. If $j = 0$, then I contains a linear form and we can conclude as in the first paragraph. Thus $j \neq 0$ and $\dim_\kappa(I/J) = 1$ (otherwise we could find an element as above in I with $j = 0$). We conclude that I has the form $(\alpha + x^2, \beta + y^2, g + hx + iy + jxy)$ with $j \neq 0$ and has colength 3. In this case $a + dx^2 + y^2 \in I^2$ is impossible. This can be shown by a direct computation, but we prefer to argue as follows. Namely, to prove this statement we may assume that κ is algebraically closed. Then we can do a coordinate change $x \mapsto \sqrt{\alpha} + x$ and $y \mapsto \sqrt{\beta} + y$ and assume that $I = (x^2, y^2, g' + h'x + i'y + jxy)$ with the same j. Then $g' = h' = i' = 0$ otherwise the colength of I is not 3. Thus we get $I = (x^2, y^2, xy)$ and the result is clear. $\square$